

T3N Invoice Approver — Enterprise Agent

Submission

Track: Useful enterprise agents on Terminal 3, easy to maintain post-challenge.

Repo (public): <https://github.com/Hud-sonn/t3n-invoice-approver>

Agent DID: `did:t3n:3281...bdc` **Contract:** `z:3281...:invoice-contracts` **ID 987**

v0.4.1

Card:

`https://cn-api.sg.testnet.t3n.terminal3.io/api/agent-card/did:t3n:3281e12c1580a44fb0a5133e5522eb1a4e180bdc`

Running choice: Continue running via startup program/listing; handover-ready (see §6).

1. What it does

`review-invoice` (no PII) validates totals, vendor allowlist, spending limits. `approve-payment` charges via Stripe test mode with PII resolved host-side through `http-with-placeholders` — bank keys never enter prompts, logs, or WASM memory. Secrets live in private KV `z:<tid>:secrets`; single egress host; every dispatch lands in the audit log. Finance teams approve invoices without pasting sensitive data into AI context.

2. Why it is maintainable

- Thin TEE: 2 functions, ~200 lines Rust, same envelope + WIT pattern as reference.
- Policy in one `config.ts` (limits, allowlist, hosts) — no Rust change for policy tweaks.
- Single-version pin (`0.4.1`), recorded `contract_id`, re-grant ACLs on bump.
- Repeatable: `docs/LOCAL_TEST.md`; weekly ops ~30 min (credits, logs, delegation windows, SDK changelog).
- Stack: TypeScript/Node/Express/Postgres-mirror — no exotic deps.

3. How to run

```
npm install
```

```
npx tsx --env-file=.env app/quickstart.ts
```

```
# Connected as: did:t3n:... / TenantClient ready.
```

Keys in gitignored `.env` / `.env.agent` only. Never commit or screenshot full keys.

4. Screenshots (paste 6 terminal shots, keys redacted)

Shot 1 — Auth: quickstart → Connected as + TenantClient ready.

[PASTE IMAGE]

Shot 2 — Build: cargo build Finished + z_tenant_flight.wasm 199K.

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Shot 3 — Register: Registered invoice-contracts contract_id 987 + contract.json.

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Shot 4 — Maps: secrets/invoices/audit ok + ACLs → 987 + entryGet present (value

hidden).

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Shot 5 — Grant + invoke: Self-grant written + HTTP 401 (caption: proves egress, fails

closed on placeholder).

[PASTE IMAGE]

Shot 6 — Card: host-card URL + curl serving agent-card.json (Invoice Approver).

[PASTE IMAGE]

5. Bugs faced

D a t	St	Symptom	Cause / Fix
	e		

e	p		
0			
9	a	Missing	.env not auto-loaded → <code>npx tsx --</code>
-	ut	T3N_API_KEY	<code>env-file=.env</code>
1	h		
1			
0			
9	in		Split install with <code>--no-audit --no-</code>
-	st	npm timeout 120s	<code>fund (21s)</code>
1	all		
1			
0			
9	cl		
-	o	clone timeout	<code>--depth 1</code> after <code>ls-remote</code>
1	n		
1	e		
0			
9	ru	system rustc 1.75,	rustup stable 1.98.1 + <code>target add</code>
-	st	no wasip2	<code>wasm32-wasip2</code>
1			
1			
0			
9	te	wasm exec perm	<code>config pins wasm → --target</code>
-	st	denied	<code>x86_64-unknown-linux-gnu (7</code>
1			<code>pass)</code>
1			
0	m	stale ACLs after	<code>create then update { only:</code>
9	a	re-register	<code>[987] };</code> control-plane entrySet

-			
1	ps		bypasses ACL
1			
0			
9	in	HTTP 401 on	Not a bug — proves
-	vo	placeholder	grant+egress+TEE; replace with real
1	ke		test key for demo
1			
0			
9	ca	registry shows	
-	rd	agent:(none), curl	URI vs body steps; curl endpoint is
1		serves card	proof
1			

6. Handover / running

Prefer **continue running** (startup program/listing). Handover if required: rotate tenant + Stripe keys in `secrets`, bump version + re-register, re-grant map ACLs + `updateMemberDelegation`, fresh `AGENT_KEY` + re-host card, revoke old grants (`grants: []`), one test approve on testnet. Full steps in `docs/HANDOVER.md`.

7. Links

- Repo: <https://github.com/Hud-sonn/t3n-invoice-approver>
- Claim: <https://go.terminal3.io/adk-community>
- Docs: quickstart, set-up-dev-env, write/build/register/invoke-contract, register-agent, member-delegation, common-errors, ai-coding-assistants, llms.txt (docs.terminal3.io)

Bundled in repo: `app/` (quickstart, register, setup-maps, delegate, whoami-agent), `z-tenant-invoice/`

(Rust source + WIT), agent-card.json, docs/. Left on PC (gitignored): node_modules/, target/,

*.wasm, .env, .env.agent. Deps: @terminal3/t3n-sdk@5.2.0, tsx (see package.json).